

# All About Transferring Power Wirelessly

Battery technology can be said to be at its peak now. So it is high time that alternative sources of charging came into play. This article focuses on wireless power transfer technology and the opportunities provided by it

ASHWIN GOPINATH

One of the biggest challenges before consumer electronics makers today is to increase the battery life of their devices on a single charge. As more powerful processors, larger memory and extra sensors are added to devices for improved performance, these consume far more power than their predecessors. Improving battery technology is one way to counter this problem, but advances in battery technology seem to have peaked now.

The only option left is to look at alternative sources of charging a device without causing inconvenience to the user. Wired chargers are a hassle for most users, and that's where the novel idea of utilising wireless charging came up.

## What's wireless power transfer?

Simply put, wireless power transmission refers to a power source that supplies a system with energy without the hassle of interconnecting wires. Wireless power transmission problems differ from those of wireless telecommunications, such as radio. While in wireless power efficiency is considered to be the more significant parameter, in wireless telecommunications the proportion of energy received becomes critical only if it is too low for the signal to be distinguished from the background noise.

There are more than a couple of technologies that enable wireless power transfer, each having its own merits and demerits.

**Coupling.** It is defined as the transfer of electrical energy from one part of the circuit to another. For example, en-

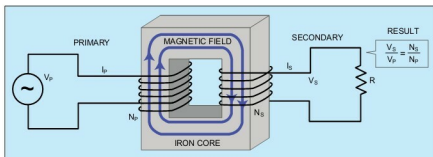


Fig. 1: Coupling

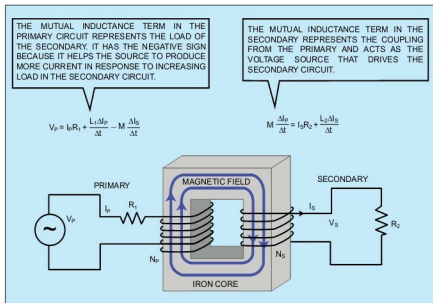


Fig. 2: Mutual inductance

ergy can be transferred from a power source to an electrical load by means of conductive coupling, which may be either hard-wire or resistive. Also, AC potential may be transferred from one segment to another (having a DC potential) using a capacitor.

**Mutual inductance.** It is defined as the phenomenon by which a change of current in a coil induces an emf in another coil placed near the first coil. The primary coil is the one in

which current changes, while the coil in which emf is induced is called the secondary coil.

**Capacitance.** Capacitance is the ability of a body to store an electrical charge. Any object that can be electrically charged exhibits capacitance. A common form of energy storage device is parallel-plate capacitor. In a parallel-plate capacitor, capacitance is directly proportional to the surface area of the conductor plates and inversely pro-